Practical 10: Volcanism and explosive activity

On successful completion of this practical, you should be able to:

- Recognise and describe the characteristic textures of pyroclastic rocks in hand specimen and thin section
- Interpret the deposition mechanism of pyroclastic rocks using evidence from observations
- Use an isopach map to reconstruct the VEI of an explosive volcanic eruption
- Discuss the challenges and uncertainties of quantifying explosivity for ancient eruptions

Part 1 (20 minutes): Hand specimens and thin sections of pyroclastic rocks

16684 or 125 *Taupo, New Zealand*

These samples are from an ignimbrite deposited by a pyroclastic density current during the AD 186 Taupo eruption, New Zealand.

Look at the hand specimens and describe or sketch the texture. Can you identify pumice clasts and lithic fragments? Do you think these are welded or unwelded ignimbrites?

Now look at the thin sections. The 'B' sections have been cut perpendicular to the lamination in the hand specimen, and the 'A' sections have been cut parallel to the lamination. Examine both thin sections, paying particular attention to the textures.

Make a labelled sketch of the key textural features.

Check that you can identify the following key features and textures:

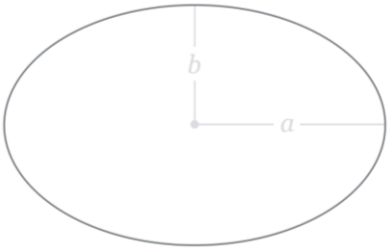
- The rock comprises pumice clasts, lithic fragments, crystals, glass shards.
- The groundmass is composed of flattened and welded glass shards.
- Pumice clasts may be welded: squashed while still hot into dark glassy fiammé.
- The rock has a eutaxitic texture. The degree of eutaxitic deformation (that is, compaction and welding) tends to be greatest towards the base of a flow, where the pressure of the overlying flow is greater and the heat is retained for longer.
- Lithic fragments are often dense and crystalline, and may have a different mineralogy to the groundmass.
- The phenocryst assemblage can be used to attempt a petrological classification. In these samples you should find essential minerals quartz, alkali feldspar (sanidine) and plagioclase; you may also find examples of type minerals augite, hornblende, biotite; accessory magnetite. Many phenocrysts are rounded or fractured.

Part 2 (40 minutes): Reconstructing a volcanic eruption from its deposits

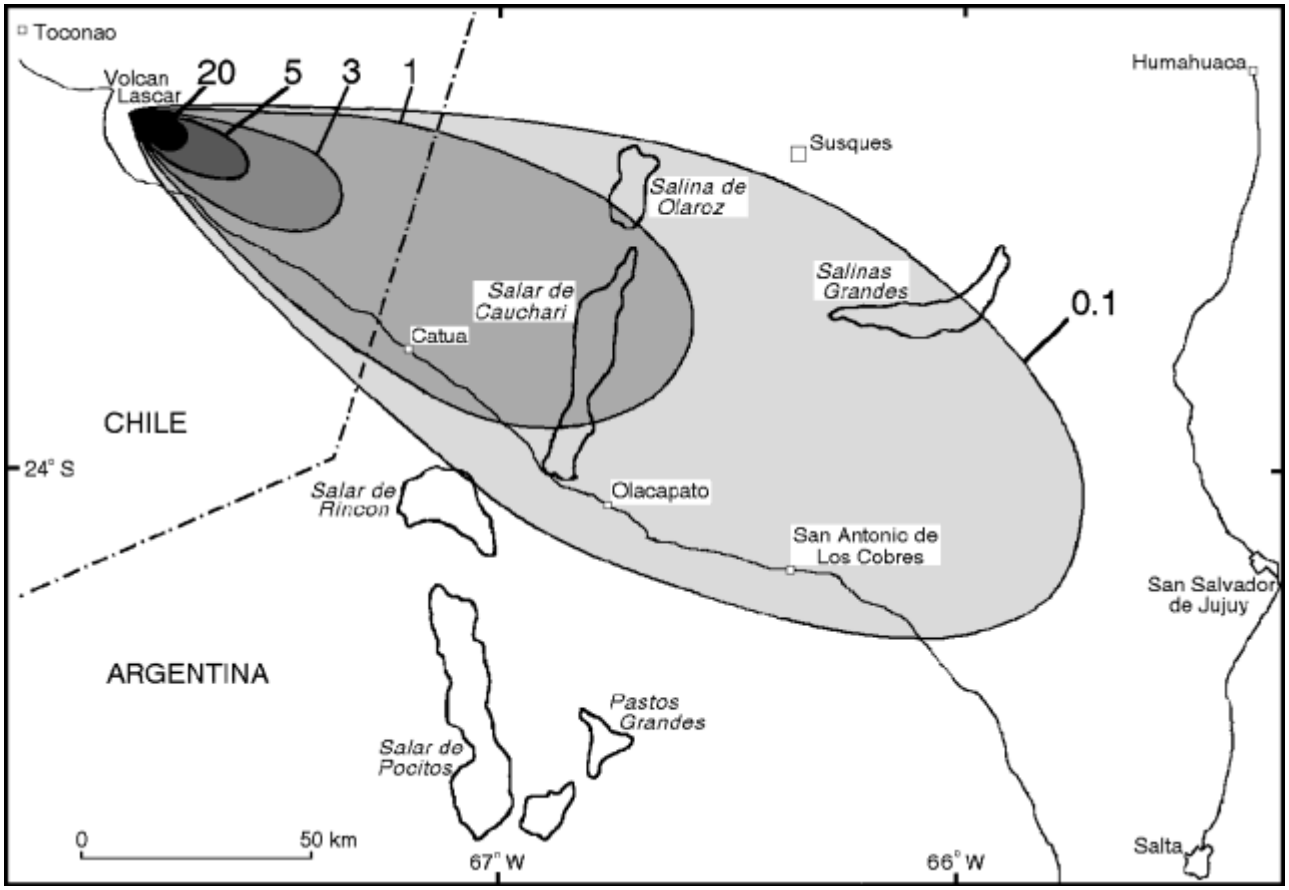
Volcan Lascar is the most active volcano of the northern Chilean Andes. It typically erupts basaltic andesitic, andesitic and dacitic magmas, and produces a range of eruption styles including lava effusion, dome growth, and explosive eruptions generating pyroclastic falls and pyroclastic density currents. The last recorded eruptive activity at Lascar was in April 2017. The largest historical eruption (i.e. observed and recorded by humans) took place in 1993.

The figure below shows an isopach map for the April 1993 eruption of Lascar. Use the isopach map to calculate an estimate of the erupted volume of pyroclastic material from this eruption. You can use the table overleaf to assist with your calculations. You may wish to work in pairs or groups and divide the calculations between you.

Hint 1: Simplify the problem by making the geometric assumption that each isopach is an ellipse. The formula for the area of an ellipse $A = \pi ab$, where a and b are the semi-major and semi-minor axes.



Hint 2: Make sure you use the same units for all the measurements in your calculation.



Isopach map for the April 1993 eruption of Volcan Lascar. Isopach thicknesses are given in cm.

Tephra thickness of isopach	Semi-major axis of isopach	Semi-minor axis of isopach	Area of isopach	Tephra volume of isopach
			Total erupted volume	

I did the calculation in Excel; here is a copy of my table. I measured the major and minor axes of the ellipse for each isopach, and then calculated the semi-major and semi-minor axes, the area, and the volume. Allow a margin of uncertainty for your measurements. Given the geometric approximations made in calculating the isopach area, there is little value in reporting the erupted volume to any more than two or three significant figures.

Isopach, cm	Isopach, km	Major axis, km	Minor axis, km	Semi-major, km	Semi-minor, km	Area, km^2	Volume, km^3
20	0.0002	14	7	7	3.5	77	0.015
5	0.00005	28	10	14	5	220	0.011
3	0.00003	50	19	25	9.5	746	0.022
1	0.00001	129	53	64.5	26.5	5370	0.054
0.1	0.000001	222	91	111	45.5	15867	0.016
					Total erupted volume		**0.118**

Use your calculated erupted volume to estimate the VEI of the 1993 Lascar eruption. You will need to refer to the VEI table in the lecture notes.

The erupted volume is between 0.1 and 1 km^3, which makes this a VEI 4 eruption.

Use the VEI table to classify the eruption style. Hence, classify the eruption style, and estimate the height of the volcanic plume.

VEI 4 eruptions are vulcanian to Plinian, and their plume heights are between 10 and 25 km.

The measured maximum plume height of the Lascar 1993 eruption was 14-16 km (although one phase of the eruption had a plume height of 10-12 km). Is your VEI classification consistent with these observations?

Yes, the observations are with the predictions made using the isopach map.

Volcan Lascar is a persistently active volcano. The eruption you have been investigating occurred in 1993. According to the VEI, when should the next eruption of a similar style and explosivity occur (or have occurred)?

VEI 4 eruptions should occur less frequently than every 1 year, but more frequently than once every 10 years. This means that we should have observed the next VEI 4 eruption at Lascar some time before 2003.

Since the 1993 eruption, Lascar has produced two VEI 3 eruptions (in 2005 and 2006), seven VEI 2 and three VEI 1 eruptions. There has not been a VEI 4 eruption since 1993. In light of this information, comment on whether the VEI is a reliable indicator of eruption frequency.

The pattern of historic activity at Lascar does indicate that there is an inverse correlation between eruption frequency and eruption magnitude – smaller eruptions happen more frequently. However, the style and frequency of activity is actually somewhat unpredictable. Eruptions of the same magnitude are not necessarily evenly spaced in time, and the repose period between eruptions of the same VEI is not necessarily the same as predicted by the VEI. This is particularly true at the higher end of the VEI scale.

Critical analysis: The VEI is very useful for quantitative classification of recent eruptions where deposit thicknesses can rapidly be measured in the field, and eruption column height can be measured using ground- or satellite-based observations. What problems might arise when trying to reconstruct the VEI of an ancient volcanic eruption, and why?

For an ancient volcanic eruption, the only information available to reconstruct VEI is the deposits preserved in the geological record. We need to know the distribution and thickness of the deposits – i.e., we need a good isopach map. So the limiting factor in reconstructing the VEI is whether or not the deposits are preserved in the geological record so that a reliable isopach map can be constructed. This presents numerous challenges, for example:

- If most of the deposits landed in a marine, lacustrine or glacial environment, they may be inaccessible.
- All pyroclastic deposits will be susceptible to erosion. Thin, distal deposits will be more susceptible to complete removal than thick, proximal deposits.
- Volcanic deposits could be disturbed by later faulting and/or tectonic activity.

Disturbance or removal of deposits will lead to the volume of erupted material being underestimated. Depending on the scale of removal, this could lead to the VEI being underestimated.

The uncertainties and limitations of isopach maps are well understood by field volcanologists, and in some instances they can be corrected for. Despite the known limitations, VEI is still a useful tool for estimating the magnitude and explosivity of ancient eruptions, and this in turn is useful for predicting the style and magnitude of future explosive activity that could occur at the source volcano (if it is still active).

Part 3 (60 minutes): Self-directed study

The remainder of the practical class is set aside for you to practice your microscopy skills in preparation for the practical test. The thin sections from practicals 1-10 will be available to help you revise and consolidate your knowledge.

Bonus digital resources

(a) Hand specimens of pyroclastic rocks

Here are some 3D renderings of hand specimens deposited by explosive volcanic eruptions. They have very different appearances and textures.

Berufjörður ignimbrite, eastern Iceland

How many different clast types can you distinguish? Describe the shape of the clasts.

At least four clast types: Large greenish-coloured clasts; medium-sized white clasts; small near-black clasts; rare red-coloured clasts; grey clasts; purplish clasts. It is hard to tell what each clast type is, but from the labels it appears that the dark-coloured clasts are likely to be lithic fragments, and the large pale green clasts are likely to be pumice fragments. All the clasts are set in a fine-grained light grey matrix. The clasts are angular and irregular shapes.

How well sorted are the clasts? Is there any indication of bedding?

Very poorly sorted. There is a possible indication of bedding, but there is no clear indication of normal or reverse grading.

What kind of pyroclastic deposit might this specimen have come from?

The poor sorting tells us this must be a pyroclastic density current deposit. It's not likely to be a surge deposit, because there is no evidence of sedimentary structures. It's not likely to be a block-and-ash deposit – the clasts in a block-and-ash deposit would be strongly skewed towards dense lithic fragments. The most probable deposit type is an ignimbrite. This comes from somewhere within flow unit 2, but the specimen is too small to see whether the lithics are normal or reverse graded.

Welded ignimbrite, Gran Canaria

How many different clast types can you distinguish?

At least four clast types: near-black, mid-grey, light grey, yellowish. All the clasts are set in a very fine-grained beige matrix.

Comment on the shape and texture of the clasts, and suggest how this texture might have formed.

Most of the clasts are flattened and elongated, but one or two are sub-rounded. The sub-rounded clasts are likely to be dense lithic fragments. The flattened clasts are likely to have been vesiculated pumices. This sample has a classic fiamme texture, which forms when still-hot pyroclasts undergo ductile deformation by flattening and compaction due to the weight of the overlying deposit.

What kind of pyroclastic deposit might this specimen have come from?

This is a welded ignimbrite, so it was deposited in a pyroclastic density current.

Compare and contrast: Which of the two specimens comes from a hotter pyroclastic deposit? What factors might influence the temperature within a pyroclastic deposit?

The welded ignimbrite must have been hotter because the pumices have undergone ductile deformation. The Iceland ignimbrite does not appear to have undergone any significant welding or compaction, so the

clasts were probably not so hot, nor the deposit so thick, as the Gran Canaria ignimbrite. As a general rule we would expect the hottest and thickest deposits to be found closer to the volcanic vent. Therefore, proximal ignimbrite deposits are more likely to weld, and distal ignimbrite deposits are more likely to retain their original depositional texture.

(b) Virtual outrcops of pyroclastic deposits

Here are some virtual outcrops of pyroclastic deposits. The digital labels are useful for identifying the main features of the volcanic deposit. You can use the virtual outcrops to practice making observations of pyroclastic rocks and interpreting the volcanic processes that formed the deposit.

Ignimbrites, Rambla de las Negras, Cabo de Gata, Spain

Notes: A debris avalanche deposit is another term for a block-and-ash deposit. Fumarolic pipes are gas escape pipes; they form when there are trapped gases within the ignimbrite flow. The trapped gases could come from still-degassing pumices, or from vapourised groundwater.

Key observations:

The most obvious feature of this outcrop is the spectacular reverse grading. The sorting is poor. The sharp basal contact and erosive top contact are both visible: this flow unit was deposited on top of a pale white ignimbrite deposit, and is overlain by a debris avalanche deposit. The predominant clast type appears to be light grey pumices. There is a huge range of clast types, and the range of clast types appears to with height. There are some fumarolic gas pipes to the right of the outcrop which appear to be well consolidated and have a dark yellow-brown colour.

The debris avalanche deposit is very coarse and poorly sorted. The basal contact is irregular and erosive. There is some evidence for reverse grading, with finer material being found immediately above the base of the bed. There is possibly some weak indication of clast imbrication.

Interpretation:

This is an ignimbrite deposited by a clast-rich pyroclastic flow, most probably caused by a collapsing eruption column. The style of eruption was therefore probably vulcanian or Plinian – it can't have been a Hawaiian or Strombolian eruption because these styles do not produce eruption columns. The fumarolic pipes could indicate that the pumices were still in the process of degassing and vesiculating when they were deposited – in which case, they must still have been quite hot when they were deposited – although not hot enough to weld. The clast size and unit thickness suggest that this outcrop represents a high-energy density current. This could mean that the outcrop is located relatively proximal to the vent and/or that the deposit represents quite a large eruption (VEI 3+).

The debris avalanche deposit was deposited by a block-and-ash flow. This could have been generated by subsequent collapse of a volcanic dome (a 'hot' debris avalanche) or else by the collapse of a crater wall or unstable slope on the volcano (a 'cold' debris avalanche). Clast imbrication suggests that the flow direction could have been right to left on the outcrop, although the observational evidence for this is not strong.

Pyroclastic surge deposits, Playa de los Genoveses, Almeria, Spain

Focus on the fresh pyroclastic deposits, rather than the altered deposits on the left of the virtual outcrop.

Key observations:

There are several different pyroclastic units in this outcrop. The lowest unit is a poorly sorted unwelded ignimbrite deposit. The clasts are angular and irregular; they mainly comprise grey-white pumices in a fine grey matrix; there are also dark grey to black clasts that are probably lithic fragments. The unit appears to be massive and poorly bedded; there is some indication of reverse-graded pumices but this is not very clear. The second unit is a fine-grained laminated deposit displaying truncated cross-bedding. Its basal contact is irregular and erosive. The finely laminated beds contain some large, angular clasts but overall the sorting appears to be moderate to good. This unit is cut by numerous sub-vertical faults with small displacements. The second unit is overlain by another finely laminated unit that is darker grey in colour. There is no evidence for truncated cross-bedding in this upper unit. The grain size is fine, so it is difficult to observe and comment on the sorting or grading in these beds.

Interpretation:
The three units probably represent three different eruptive events. The lowest unit was deposited by a clast-rich pyroclastic flow, probably generated by a collapsing eruption column.
The second unit is a pyroclastic surge deposit; the evidence for this is the fine lamination and the sedimentary structures like truncated cross-bedding. The surge deposit represents a dilute pyroclastic density current – these are possibly produced when conditions in the eruption column are very close to the boundary condition that separates stable convection from column collapse.
The uppermost unit is labelled as an ash fall deposit, presumably because the thin individual beds are normally graded. If it is a fall deposit then it must represent pyroclastic fallout from a stable convecting eruption column. However, the deposit does not appear to be laterally continuous, nor to mantle the topography of the underlying pyroclastic surge deposit – so perhaps it represents a second pyroclastic surge. An alternative possibility is that the unit has been disturbed by post-depositional faulting that destroyed the lateral continuity of the fall deposit.